

DOP :

DOS :

NLP PRACTICAL MANUAL - Prof. D V Chandran

EXPT 9 : Implementing Named Entity Recognition (NER) using NLTK , SpaCy and Stanford NER in Python AND using INLTK for regional language.

AIM : WAP to implement Named Entity Recognition (NER) using NLTK , SpaCy and Stanford NER in Python AND using INLTK for regional language.

THEORY : Named entity recognition is a natural language processing technique that can automatically scan entire articles and pull out some fundamental entities in a text and classify them into predefined categories. Entities may be,

- Organizations,
- Quantities,
- Monetary values,
- Percentages, and more.
- People's names
- Company names
- Geographic locations (Both physical and political)
- Product names
- Dates and times
- Amounts of money
- Names of events

Named Entity Recognition is the process of detecting the named entities such as person names, location names, company names, etc from the text. It is also known as *entity identification* or *entity extraction* or *entity chunking*.

Person

Product name

Location

e.g. **John** wants to purchase a new **Samsung** phone from **Chennai**.

e.g.

London is the capital and most populous city of **England** and the **United Kingdom**.
Geographic Entity Geographic Entity Geographic Entity

Steps in NER :

1. Noun phrase identification - John , India , Google , **Zoom**
2. Phrase Classification - Person , Location , Organisation , **Person**
3. Entity Disambiguation - Zoom - Organisation

Use cases of NER :

1. Customer Support
2. Recommendation System
3. Resume Summarisation
4. Automated Chatbots
5. Content Analyzers
6. Consumer Insights, etc.

ALGORITHM :

INPUT - OUTPUT :

INPUT : “*John working at IBM wants to buy a new Samsung phone from Bangalore.*”

OUTPUT : John - Person
 IBM - Organisation
 Samsung - Product name
 Bangalore - Location

NOTE : *Make a comparative study of all the three libraries : NLTK , spaCy , Stanford NER / and for regional language (HINDI & Mother tongue) using INLTK & Stanza*

CONCLUSION :

PROGRAM AND OUTPUT PRINTOUT